Interaction support in smooth-number orders and corner-cut fans

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Abstract

Order the nonnegative integer lattice by a positive weight vector. How many coordinates must a comparison involve before it can first affect a prefix of length n? We prove that the minimum first tied rank of a generic comparison wall with support s is $\Theta((3\sqrt{3}/2)^s/\sqrt{s})$, with absolute constants. Consequently the worst-case coordinate-face dimension needed to reconstruct a labeled n-term ordered prefix is

$$\frac{\log n + \frac{1}{2}\log\log n}{\log(3\sqrt{3}/2)} + O(1).$$

The same parameter is the maximum support of a positive-weight facet of a prefix cone and of a sufficiently generic n-point ideal's state fan, uniformly over ambient dimension. It yields sparse linear certificates for a realizable prefix. Finite prime approximation transfers the face-information result to smooth-number exponent orders. The general Hilbert-zonotope framework has a different support scale: arbitrary length-n quotient algebras admit walls with $2\log_2 n + O(1)$ coordinates. We give an explicit boundary example and separate the structural statements from running-time claims.

1 The question and its relation to exponent streams

Let $d, n \geq 1$, and let $w \in \mathbb{R}^d_{>0}$ have rationally independent coordinates. Write $W_n(w)$ for the first n vectors of $\mathbb{Z}^d_{\geq 0}$, in increasing order of $w \cdot x$, with the zero vector at rank 1. For distinct primes $p_1, \ldots, p_d$, taking $w_i = \log p_i$ gives the exponent vectors of the fixed-basis smooth numbers. Coordinates retain their labels; numerical weight order is not additional input. In the prime model the labels may have any fixed order, and the numerical prime values are not supplied as additional data in the reconstruction problem.

For a nonempty coordinate set S, the *face word* $W_n(w_S)$ orders the lattice on S. It contains n vectors from that face, not the first n entries of a coordinate projection of the full word. Let $R_{\mathrm{all}}(n)$ be the least integer r such that all labeled face words of length n on at most r coordinates determine $W_n(w)$, uniformly in d and w. Let $R_{\mathrm{prime}}(n)$ restrict the realizing weights to distinct prime logarithms. Define both parameters to be 0 at $n = 1$.

This question comes from the deletion and occurrence structure of the individual exponent streams. If $E_i(j)$ is coordinate i of the j-th vector, define $T(E_i)(j)$ as the number of earlier occurrences of the same symbol. Fixing coordinate i leaves the order on the other coordinates. Therefore

$$T(E_i)(j) = \rho_i(x_j \text{ with coordinate } i \text{ deleted}) - 1, \qquad T^2(E_i) = E_i,$$

where ρ_i is the one-based rank on the complementary face. Indeed occurrences of each face rank have clock exponents $0, 1, 2, \ldots$. Deleting zero entries of E_i and subtracting one corresponds

to translating lattice points by the i-th unit vector, and reproduces the full order. Placement arrays and signature-sequence duality are established background [8, 9]. For two generators the partner is the other exponent; in higher dimension it encodes the rank of an entire face vector. Our concern is the finite information needed to recover the ordering hidden in these ranks.

Put $\gamma = 3\sqrt{3}/2$. All unqualified logarithms below are natural.

Theorem 1 (Sharp face information)**.** *As $n \to \infty$,*

$$R_{\mathrm{all}}(n) = R_{\mathrm{prime}}(n) = \frac{\log n + \frac{1}{2}\log\log n}{\log \gamma} + O(1).$$

The equality of the two parameters is exact for each n; the displayed asymptotic determines their value to bounded additive error. Constants are independent of ambient dimension and of the prime bases.

The statements concern ordinal exponent data. An arithmetic algorithm supplied with the primes can compare numerical products directly. Nor is requesting all faces cheap: their number may be $\sum_{j \le r} \binom{d}{j}$, each containing n vectors. No random-access running-time improvement follows from Theorem 1 alone.

2 Comparison walls and a composition estimate

For a primitive mixed-sign $z \in \mathbb{Z}^d$, put $u = z_+$ and $v = z_-$. A *generic positive wall point* is $w > 0$ such that

$$\{a \in \mathbb{Z}^d : w \cdot a = 0\} = \mathbb{Z}z.$$

At height $h = w \cdot u = w \cdot v$, the only tied points are u, v. To see this, a point at that height differs from u by an integer multiple of z; nonnegativity permits only the two endpoints. The lowest tie anywhere in the lattice is this pair: every other tied pair is a nonnegative translate of positive integer multiples of u, v. Its *first tied rank* is $1 + |\{x : w \cdot x < h\}|$.

Let $J(s)$ be the minimum first tied rank of a generic positive wall whose normal has support s. One may work in dimension s: additional coordinates can only increase the count, and can be chosen large when proving an upper bound.

Lemma 2 (Box bound)**.** *At a generic positive wall,*

$$J \ge \frac{1}{2}\prod_i (1 + |z_i|) \ge 2^{s-1}.$$

Proof. Pair the points x and $u + v - x$ in the integer box $0 \le x \le u + v$. Their scores add to $2h$, and only u, v have score h. Thus exactly $(\prod_i (1 + |z_i|) - 2)/2$ box points lie strictly below h. The product is even because a primitive vector has an odd coordinate. □

The box and flip viewpoints are classical ingredients [5, 6]. The additional points with repeated coordinates are essential for the sharper bound.

Lemma 3 (Strict composition halfspaces)**.** *Let $s \ge 2$, $q = \lfloor s/2 \rfloor$, and let X be uniform on $\{x \in \mathbb{Z}^s_{\ge 0} : \sum_i x_i = q\}$. If $\beta \ne 0$ and $\sum_i \beta_i = 0$, then, for $Y = \beta \cdot X$,*

$$\mathbb{E}Y^4 \le 12(\mathbb{E}Y^2)^2, \qquad \Pr(Y < 0), \Pr(Y > 0) \ge 1/48.$$

Proof. Use falling factorials $(q)_j$, rising factorials $s^{(j)}$, and $u_j = (q)_j/s^{(j)}$, interpreted as zero for $j > q$. The standard factorial-moment identity is

$$\mathbb{E}\prod_i (X_i)_{a_i} = \frac{(q)_{\sum a_i}\prod_i a_i!}{s^{(\sum a_i)}}.$$

For a self-contained derivation, the numerator's generating function is

$$\prod_i (a_i!)t^{\sum a_i}(1-t)^{-s-\sum a_i}.$$

Take its coefficient of t^q and divide by $\binom{s+q-1}{q}$. This is also the all-ones-parameter Dirichlet–multinomial specialization of the published general formula [7, Section 4.2 and Appendix A].

Put $A = \sum_i \beta_i^2$ and $C = \sum_i \beta_i^4 \le A^2$. Expanding powers with the factorial moments and $\sum_i \beta_i = 0$ gives

$$\begin{aligned}\mathbb{E}Y^2 &= (u_1+u_2)A,\\ \mathbb{E}Y^4 &= (u_1+7u_2+12u_3+6u_4)C + (3u_2+6u_3+3u_4)A^2.\end{aligned}$$

Set $r = q/s$ and $t = u_2$. Then $1/3 \le r \le 1/2$, $0 \le t \le r^2$, $u_3 \le rt$, and $u_4 \le r^2 t$. Hence

$$\frac{\mathbb{E}Y^4}{A^2} \le r + (10+18r+9r^2)t \le 12(r+t)^2.$$

For the last inequality, the difference between right and left is decreasing in t on the indicated interval: its derivative is at most $6r + 15r^2 - 10 < 0$. At $t = r^2$ the difference is $r(2r + 6r^2 + 3r^3 - 1) > 0$.

Write $\sigma^2 = \mathbb{E}Y^2 > 0$. Hölder interpolation gives $\mathbb{E}|Y| \ge \sigma^3/\sqrt{\mathbb{E}Y^4} \ge \sigma/\sqrt{12}$. Since $\mathbb{E}Y = 0$, $\mathbb{E}Y_- = \mathbb{E}|Y|/2$. Cauchy–Schwarz bounds this by $\sigma\sqrt{\Pr(Y<0)}$, giving $1/48$. Apply the same argument to $-Y$. □

No separate novelty claim is made for the factorial moments, the fourth-moment method, or the particular nonoptimal constant.

Lemma 4 (Strict lattice count)**.** *If $w_i > 0$, $h > 0$, and $\sum_{i=1}^s w_i \le 2h$, then*

$$|\{x \in \mathbb{Z}^s_{\ge 0} : w\cdot x < h\}| \ge \frac{1}{48}\binom{s+\lfloor s/2\rfloor - 1}{\lfloor s/2\rfloor}.$$

Proof. Put $q = \lfloor s/2\rfloor$. If the weights are not all equal, apply Lemma 3 to $\beta_i = w_i - \sum_j w_j/s$. At least the stated number of degree-q points are strictly below their mean $q\sum_i w_i/s \le h$. If all weights equal a, the entire layer is strictly below h unless $qa = h$. In that exception $s = 2q$; the degree-$q-1$ layer is below h and has

$$\binom{s+q-2}{q-1} = \binom{s+q-1}{q}\frac{q}{s+q-1} \ge \frac{1}{3}\binom{s+q-1}{q}.$$

This handles the zero-variance case separately. □

3 The sharp interaction scale

Theorem 5. *Uniformly for $s \ge 2$,*

$$J(s) = \Theta(\gamma^s/\sqrt{s}).$$

For every prescribed finite M, the upper constructions can be perturbed to opposite sides of a wall while preserving all proper coordinate-face words through M terms.

Proof. At a wall with support s, each nonzero $|z_i| \geq 1$, so $\sum_{i \in \operatorname{supp} z} w_i \leq w \cdot (u+v) = 2h$. Lemma 4 and Stirling's formula give the lower bound, since

$$\log \binom{s + \lfloor s/2 \rfloor - 1}{\lfloor s/2 \rfloor} = s \log \gamma - \tfrac{1}{2} \log s + O(1).$$

For the upper bound put $a = \lfloor s/2 \rfloor$, $b = \lceil s/2 \rceil$, and $z = (1^a, -1^b)$. Start with weights b/a in the first group and 1 in the second. The common height is b. Choose a generic point on this wall within $\eta < 1/(s+1)$ of these weights. Its common height is at most $b + a\eta$, whereas every point of degree at least $b+1$ has score at least $(b+1)(1-\eta) > b + a\eta$. Therefore

$$J(s) \leq \binom{s+b}{b} = \Theta(\gamma^s / \sqrt{s}).$$

Such generic points exist by avoiding the countably many other rational hyperplanes inside a relatively open wall patch.

Every proper coordinate subvector at that point is independent: a relation on a proper face cannot be a nonzero multiple of the full-support normal z. Each prescribed finite face word is determined by finitely many strict inequalities. A sufficiently small perturbation of the last weight in either direction preserves all of them. It reverses the lowest full tie, changing the full word at its first tied rank. Perturbations can be chosen off all rational hyperplanes. □

Lemma 6 (Finite prime realization)**.** *Any finite collection of strict positive-weight comparisons can be preserved by distinct prime logarithms. Two perturbations differing only in the last coordinate can share all their other prime bases.*

Proof. For large L, Bertrand's postulate supplies a prime in $(e^{Lw_i}, 2e^{Lw_i})$. The normalized logarithmic error is at most $\log 2/L$. All finitely many strict comparisons persist for large L. Choose the target vectors with distinct coordinates; their intervals become disjoint, giving distinct bases. Shared coordinates use the same intervals and primes. □

Finite prime realization is established in [10, Proposition 6.2]; the lemma records the elementary version used here, including shared coordinates for the two perturbations. It gives no new prime-distribution estimate.

4 Exact inversion and reconstruction

A finite prefix's weight region is an open polyhedral cone. Indeed impose its consecutive comparisons, positivity, and the comparison of its last point with every coordinatewise minimal excluded point. The excluded border is finite, and every excluded point dominates a border point.

Theorem 7 (Exact interaction parameter)**.** *For $n \geq 2$, let $\kappa(n) = \max\{s \geq 2 : J(s) \leq n\}$. Set $\kappa(1) = 0$. Then*

$$R_{\text{all}}(n) = R_{\text{prime}}(n) = \kappa(n).$$

Proof. Take two weights giving the same supplied face words on at most $\kappa(n)$ coordinates. The intersection of the corresponding face-prefix cones is open and convex. Join the weights by a generic path inside it. A finite arrangement suffices: every point that can appear or first tie by rank n has every coordinate at most n, since larger values give too many smaller pure powers. The path can avoid the countably many intersections of distinct rational comparison hyperplanes. The finite relevant arrangement then detects every prefix change.

If the full prefix changes, a wall with primitive normal z has its lowest tied rank at most n. Restricting to its support S cannot increase this rank, so $J(|S|) \leq n$. Its face is among the

supplied data, and crossing the wall changes that face's length-n word. This contradicts the common cone.

Conversely, a generic s-coordinate wall attaining $J(s) \le n$ has perturbations agreeing through $M = n$ on all proper faces and differing in the full prefix. Lemma 6 transfers the two finite patterns to prime bases. Thus fewer than s coordinates cannot suffice even in the prime class. The box bound makes $\kappa(n)$ finite. □

Proof of Theorem 1. The lower estimate for $J(s)$ implies that any $J(s) \le n$ satisfies

$$s \log \gamma - \tfrac{1}{2} \log s \le \log n + O(1).$$

The box bound first gives $s = O(\log n)$, yielding the stated upper bound. For the reverse inequality use the even-dimensional upper construction with $s = 2k$ and first rank at most $\binom{3k}{k}$. Stirling gives $\log \binom{3k}{k} = 2k \log \gamma - \frac{1}{2} \log k + O(1)$. Taking

$$k = \lfloor \frac{\log n + \frac{1}{2} \log \log n}{2 \log \gamma} - C \rfloor$$

for a sufficiently large fixed C proves the matching lower bound. □

5 Sparse validity cones and interval certificates

Theorem 8. *For $n \ge 2$, every positive-weight facet of a realizable length-n prefix cone has support at most $\kappa(n)$. Some cone needs this support. The same bound is sufficient for a finite linear inequality description, retaining coordinate positivity separately.*

Proof. Choose a generic positive relative interior point of a facet. Crossing it changes the prefix, so its normal z has $J(|\operatorname{supp} z|) \le n$, as above. Coordinate boundary facets are represented by positivity. In a finite halfspace description each facet has a defining normal. Retaining these normals defines the same closed cone. Every other valid normal is a nonzero nonnegative combination of them, so their strict inequalities also imply all original strict inequalities. Realizability provides the strict interior point needed here.

Conversely a wall attaining $J(s) \le n$ supplies a genuine prefix facet of support s. Any finite inequality description must contain a normal parallel to that facet; normals with smaller support cannot define it. □

An explicit sufficient cutoff follows from the proof, without computing $J(s)$. Put $p = \lfloor (s-1)/2 \rfloor$, $q = \lfloor s/2 \rfloor$, and

$$B_s = \max\left(2^{s-1},\ 1 + \lceil \binom{s+p}{p} / s \rceil,\ 1 + \lceil \binom{s+q-1}{q} / 48 \rceil\right).$$

For the middle bound, each cyclic permutation orbit of the points of degree at most p has average height strictly below h, so supplies one point below h. Thus $J(s) \ge B_s$, and $r(n) = \max\{s : B_s \le n\}$ is sufficient. One may improve this with the elementary exact values $J(2) = 2, J(3) = 5, J(4) = 10$, proved in the research artifact.

For an actual certificate, first generate a proposed word at exact positive rational weights and replay through $n + 1$ points to verify a strict prefix and boundary. Form the usual consecutive and minimal-border comparisons. Delete all nonpositivity normals above a proved sufficient support cutoff. For a rational box $w_i \in [l_i, u_i]$, $l_i > 0$, check

$$\sum_{z_i \ge 0} z_i l_i + \sum_{z_i < 0} z_i u_i > 0$$

for every retained normal z. Theorem 8 certifies the whole box, including any prime logarithms rigorously enclosed in it. It is essential to establish a strict realization first; deletion need not preserve the feasibility question for an arbitrary impossible word. This procedure reads the whole prefix and border. It makes no running-time claim against the published recognition algorithms.

6 Generic state fans and the quotient boundary

Work over $\mathbb{C}$ in the algebraic statements; normal cones use minimization. A state-fan facet means a codimension-one wall between full-dimensional positive-weight regions. For a length-n ideal, the state polyhedron and its normal fan encode its initial ideals. The published framework [1, 2] identifies the state polyhedron of a sufficiently generic n-point configuration with the corner-cut polyhedron

$$P_n^d = \operatorname{conv}\left\{\sum_{x\in D} x : D \subset \mathbb{Z}^d_{\geq 0},\ |D| = n\right\}.$$

A precise sufficient genericity condition is that every n-element subset of $V_n^d = \{v : \prod_i (v_i+1) \leq n\}$ is a quotient basis. This identification is prior art.

Corollary 9. *For $n \geq 2$, the maximum support of a state-fan facet meeting the positive weight cone, over all ambient dimensions and sufficiently generic n-point configurations, equals $\kappa(n)$.*

Proof. A corner-cut cone is refined by the cones for its internal ordered prefixes. A positive-weight wall therefore has a primitive normal whose lowest tied rank is at most n; its support is at most $\kappa(n)$.

For the reverse bound, take a wall attaining $J(s) \leq n$ at height h. Append $n - J(s)$ new coordinate weights in $(\max(h/2, h - \min_i w_i), h)$. Each contributes exactly its unit vector below h. Squares and sums with old nonzero points or other new coordinates are above h. Choose the added weights generically, preserving only the original relation. The first tied rank is now precisely n. Across this wall the two n-point cut sums differ by $v - u$, giving a positive-weight facet with support s. The published generic-configuration identification transfers it to the state fan. □

For $n = 1$ the prefix is the zero vector and there is no positive-weight facet in either setting; no maximum over an empty family of facets is required.

The corresponding statement for arbitrary quotient algebras has a different scale. Let $A(n)$ be the maximum support of a noncoordinate state-fan facet meeting positive weights, over all length-n ideals and ambient dimensions.

Proposition 10 (Boundary for arbitrary quotients)**.** *For $n \geq 2$,*

$$2\lfloor \log_2(n+2) \rfloor - 2 \leq A(n) \leq 2\lfloor \log_2 n \rfloor.$$

Proof. The upper bound is a direct consequence of the Hilbert-zonotope refinement [2, Proposition 3.2 and Theorem 3.5]. Every refining direction is a difference of two members of V_n^d, each with support at most $\lfloor \log_2 n \rfloor$. A facet of the coarser state fan must lie in one of the refining wall hyperplanes.

For the lower bound put $k = \lfloor \log_2(n+2) \rfloor - 1$, $U = x_1 \cdots x_k$, $V = y_1 \cdots y_k$, and $\ell = n - (2^{k+1} - 2)$. Let

$$S = \operatorname{div}(U) \cup \operatorname{div}(V) \cup \{t, t^2, \ldots, t^\ell\}.$$

Let K be the monomial ideal of the complement of S and set $I = K + (U - V)$. The set S has $n + 1$ members. Every variable annihilates U, V modulo K; hence their relation removes exactly one dimension and the quotient has length n.

For either leading choice of $U - V$, the monomial generators of K and this binomial form a Gröbner basis. Indeed an S-pair with a monomial generator reduces to a nonconstant multiple of the other socle monomial, in K. Monomial–monomial S-pairs vanish. Thus the two initial ideals are $K + (U)$ and $K + (V)$, separated at positive weights by $\sum_i w_{x_i} = \sum_i w_{y_i}$, of support $2k$. □

The general upper bound is credited as an immediate published consequence. The example delimits Corollary 9; it is not proposed as an independent universal-basis algorithm. A length-6 quotient already has a support-4 wall, whereas the full lattice cannot first have one before rank 10.

7 Prior-art boundary, verification, and assessment

Corner cuts, generic state polyhedra, and recognition via finite linear inequalities are established [1, 2]. The recognition theorem of [1, Section 3] allows growing dimension and uses Borel-fixedness to obtain a complete frontier description. It assumes that the staircase itself is given. Our face input does not give that staircase, and our support bound is not a new recognition-time theorem. Wagner's fixed-dimension counting and flag bounds [3] address a different parameter. The finite-chain reductions of [4, Sections 5–6] likewise use a known full chain and its front; they can retain coprime comparisons of full support. Boolean flips and minimal determining comparisons [5, 6] are credited; repeated powers are indispensable to the lattice count here.

The edge conditions of [11, Section 5] constrain the affine span of exchanged cut points. Our generic wall exchanges only two points, so that condition alone supplies no support/rank count. The weighted-average degree bounds and greedy model selection of [12, Theorems 2–7] use the same established geometry but a different objective. The lattice-point decomposition question in [13, Theorem 4, Corollary 5 and Section 4] is also distinct from facet-normal support. These are theorem-level comparisons with the inspected results, not claims that no other derivation exists.

The candidate contribution is the sharp support/rank law, its exact face/facet interpretation, and the generic-state consequence, rather than those frameworks or the moment method. The repository's bounded primary comparison records the statements inspected and remaining priority limits. This draft does not assert independent confirmation of novelty, journal acceptance, or a new prime-distribution result.

The dependency-free research checker is `scripts/run_face_reconstruction_checks.py`; its report is `results/research/face_reconstruction_checks.json`. It checks exact moments on complete composition layers, strict boundary cases, cyclic orbits, even and odd support obstructions, prime witnesses, and sparse-cone certificates. Separate exact product and pointer generators validate prime prefixes. The quotient examples check every nontrivial Buchberger pair for both leading choices. Large finite dimension brackets use exact binomial integers. These checks support the delicate steps; the uniform theorems depend on the proofs, not a fitted asymptotic curve.

This is a structural result for growing interaction dimension. Its practical algorithmic value remains to be established. The single-stream rigidity and conditional finite-mask results are preserved separately in the research note and are not needed for the main theorem.

Reproducibility. From the repository root run

```
python3 -B scripts/run_face_reconstruction_checks.py
./build.sh
python3 tests/test_smoke.py
```

The core CLI, its independent rank auditor and previous manuscripts are unchanged.

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